

Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Social Network Analysis [IT_4411]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) In the graph in Figure 1a, node B is pivotal for two pairs: the pair consisting of A and C, and the pair consisting of A and D. (Notice that B is not pivotal for the pair consisting of D and E since there are two different shortest paths connecting D and E, one of which (using C and F) doesn't pass through B. So B is not on every shortest path between D and E.) On the other hand, node D is not pivotal for any pairs. (2)
- A) (a) Construct a graph in which every node is pivotal for at least one pair of nodes. Explain your answer.

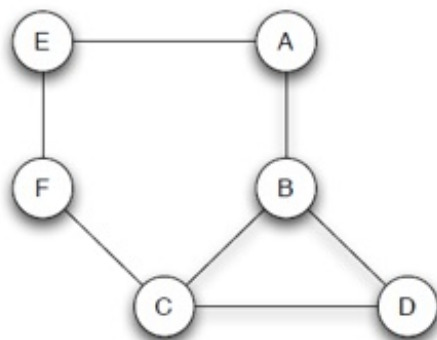


Figure 1a

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) In a large healthcare organization, a social network of doctors, nurses, and administrative staff is mapped to understand communication patterns. It is observed that: (4)
- Doctors mostly interact with other doctors.
 - Nurses communicate frequently within their own group.
 - A few nurses act as links between doctors and administrative staff.
 - There are strong triadic connections within each department (i.e., if A knows B and B knows C, then A often knows C)

Analyze how transitivity within the doctor network might enhance trust and coordination but potentially limit information diversity

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- C) (4)

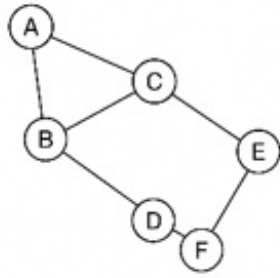


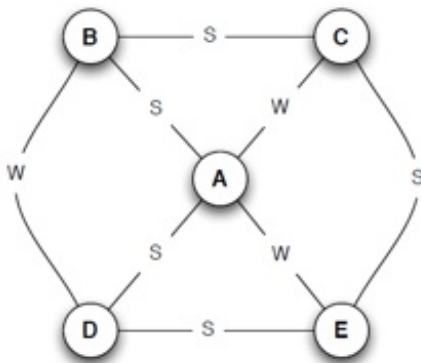
Figure 1c

Examine Depth-First Search (DFS) algorithm, traverse the graph in Figure 1c, starting from node A. List the order in which nodes are visited. Show all the steps clearly

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2) In the social network depicted in Figure 2a, with each edge labeled as either a strong or weak tie, which nodes satisfy the Strong Triadic Closure Property, and which do not? Justify your answer. (5)

A)



Course Outcome: "CO2" / Cognitive Level: "Evaluate"

- B) Given a network showing pairs of people who share activities, we can try to reconstruct an affiliation network consistent with this data. For example, suppose that you are trying to infer the structure of a bipartite affiliation network, and by indirect observation you've obtained the projected network on just the set of people, there is an edge joining each pair of people who share a focus. This projected network is shown in Figure 2b. (5)

- (i) Determine an affiliation network involving these six people, together with four foci that you should define, whose projected network is the graph shown in Figure 2b.
(ii) Justify why any affiliation network capable of producing the projected network in Figure 2b must have at least four foci.

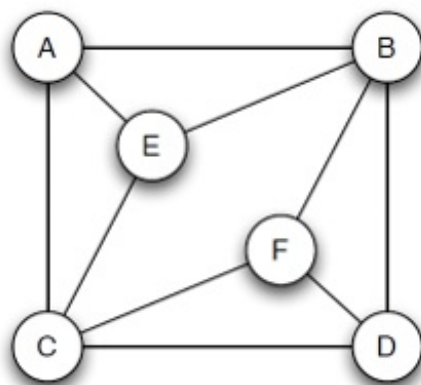


Figure 2b.

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

- 3) By Making use of social influence drives link formation in online networks, Experiment how homophily and triadic closure contribute to this process, with an example from a real-world platform. (3)

A)

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Distinguish strong and weak forms of structural balance in signed social networks. How do they help in identifying communities within complex networks? (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) Evaluate the role of Bayesian reasoning in modeling herding behavior. Explain how information cascades emerge when individuals sequentially update their beliefs. (4)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 4) Inspect threshold models and tipping points in the adoption of green technologies using Kleinberg model. (4)

A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Compare how clustering coefficient and degree distribution influence cascade robustness in Watts' threshold model. Which is more resilient to misinformation? (3)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- C) Using Granovetter's weak tie concept to Analyze innovation diffusion in inter-organizational networks, and examine its implications. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Evaluate the navigability of Kleinberg's small-world model . How does spatial probability distribution of long-range links influence search efficiency? (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- B) Distinguish the accuracy of supervised vs unsupervised link prediction models in sparse networks. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4411:Social Network Analysis

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	13
3	3	CO 3	14
4	4	CO 4	13
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4411:Social Network Analysis

#	Level	No of Questions	Marks (%)
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1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	2	5(10%)
4	Analysis	6	23(46%)
5	Create	5	22(44%)
6	Evaluate	0	0(0%)
	Total	13	50 (100%)